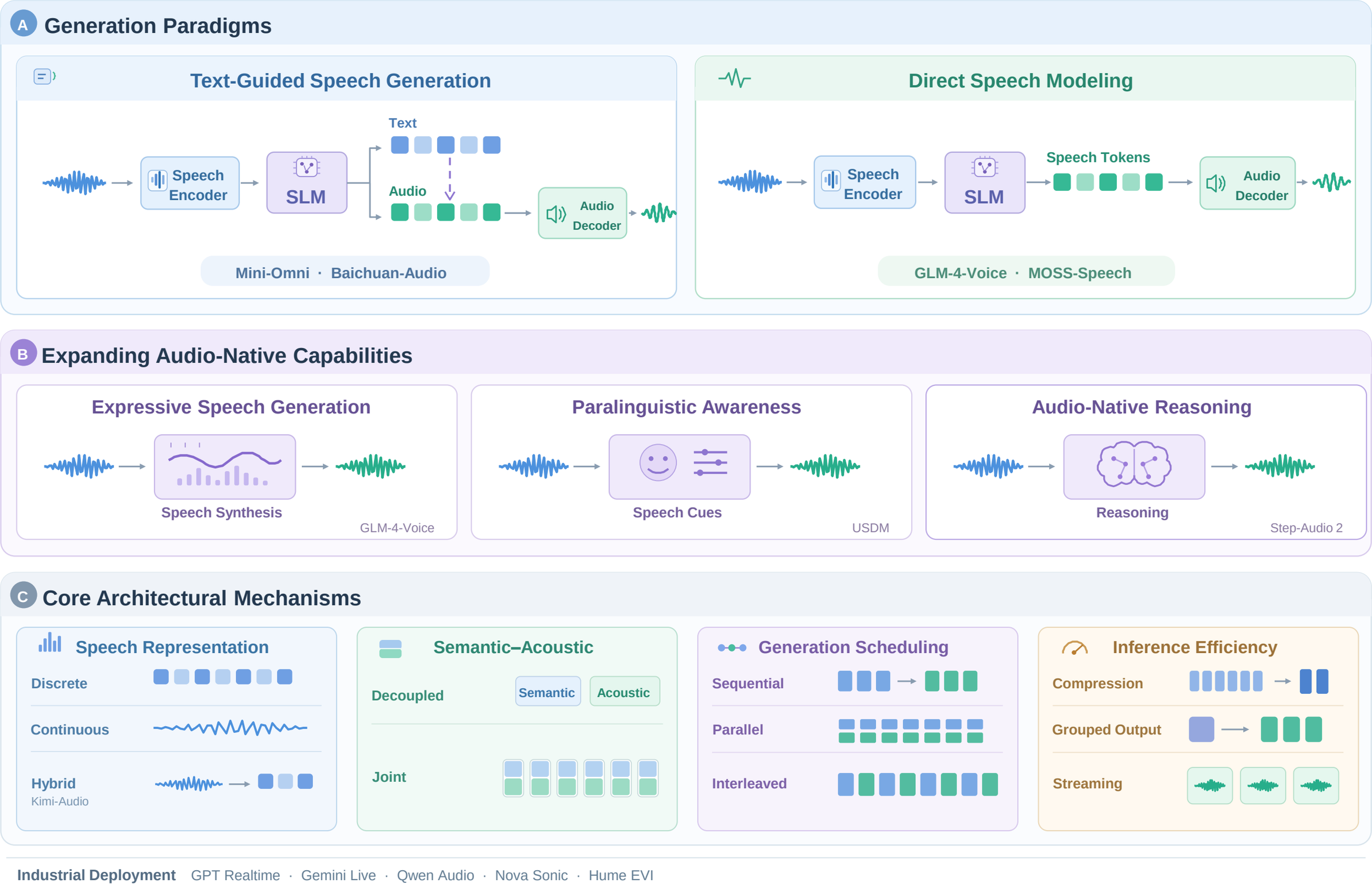

A Generation Paradigms
Text-Guided Speech Generation
Speech Encoder
SLM
Text
Audio
Audio Decoder
Mini-Omni · Baichuan-Audio
Direct Speech Modeling
Speech Encoder
SLM
Speech Tokens
Audio Decoder
GLM-4-Voice · MOSS-Speech
B Expanding Audio-Native Capabilities
Expressive Speech Generation
Speech Synthesis
GLM-4-Voice
Paralinguistic Awareness
Speech Cues
USDM
Audio-Native Reasoning
Reasoning
Step-Audio 2
C Core Architectural Mechanisms
Speech Representation
Discrete
Continuous
Hybrid
Kimi-Audio
Semantic–Acoustic
Decoupled
Semantic
Acoustic
Joint
Generation Scheduling
Sequential
Parallel
Interleaved
Inference Efficiency
Compression
Grouped Output
Streaming
Industrial Deployment GPT Realtime · Gemini Live · Qwen Audio · Nova Sonic · Hume EVI